

Chained Indices Unchained:

Structural Transformation and the Welfare Foundations of Income Growth Measurement

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Introduction

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Motivation and Research Question

- ▶ We study how to measure **welfare-relevant growth** in economies undergoing:
 - ▶ structural transformation (ST),
 - ▶ investment-specific technical change (ISTC),
- ▶ Why does this matter?
 - ▶ ST and ISTC lead to changes in:
 - ▶ relative prices,
 - ▶ expenditure shares,
 - ▶ and the composition of output.
 - ▶ These changes can generate aggregation biases in traditional quantity indices.
- ▶ We build on the economic theory of index numbers:
 - ▶ We focus on the Divisia index: the continuous-time analogue of the chained Fisher index used in national accounts.

Our Approach

- ▶ Based on the Fisher–Shell principle:
 - ▶ welfare comparisons must use a consistent preference ordering.
- ▶ Key idea:
 - ▶ compare infinitesimal changes in welfare holding current preferences fixed,
 - ▶ then chain these local welfare comparisons over time.
- ▶ This yields a **welfare-based** chained quantity index:
 - ▶ the chained Fisher–Shell quantity index,
 - ▶ which coincides with the chained Divisia index in continuous time.

$$g_t^D = s_{e,t} \left[s_{g,t} g_{c_g,t} + (1 - s_{g,t}) g_{c_s,t} \right] + (1 - s_{e,t}) g_{x,t}$$

Growth rates and nominal expenditure shares are sufficient statistics.

Main Findings

- ▶ We compare:
 - ▶ The Chained Divisia Index,
 - ▶ A Current-Base Fisher–Shell Index (EV),
 - ▶ A Reference-Base Fisher–Shell Index (CV).
- ▶ Within the model of Herrendorf, Rogerson, and Valentinyi (2021):
 - ▶ non-homothetic preferences,
 - ▶ sector-specific productivity growth,
 - ▶ investment-specific technical change.
- ▶ Quantitatively:
 - ▶ fixed-base measures systematically distort historical welfare growth,
 - ▶ with biases that grow over time under ST and ISTC.
 - ▶ fixed-base measures require periodic revisions of historical growth.
- ▶ The Divisia index only requires expenditure shares and expenditure growth rates.
- ▶ EV and CV additionally require knowledge of deep preference parameters.

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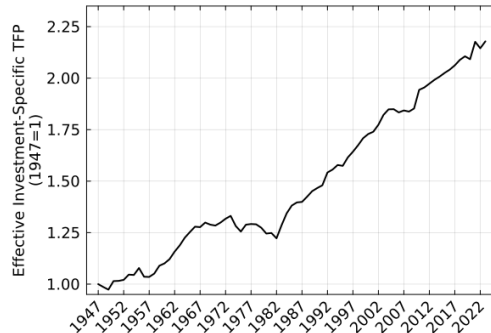
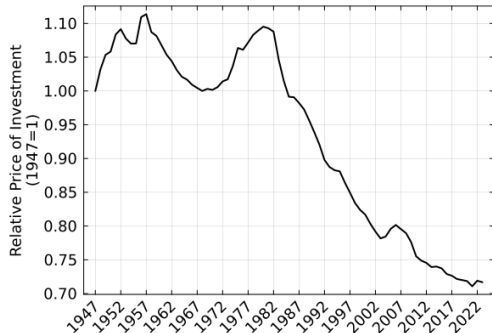
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ST and IBTC

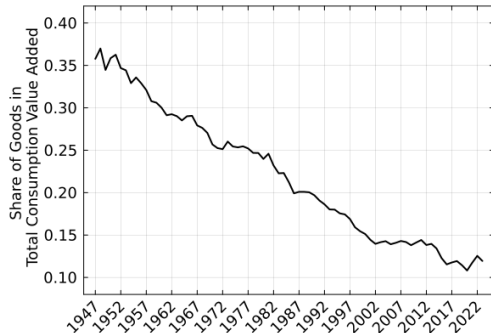
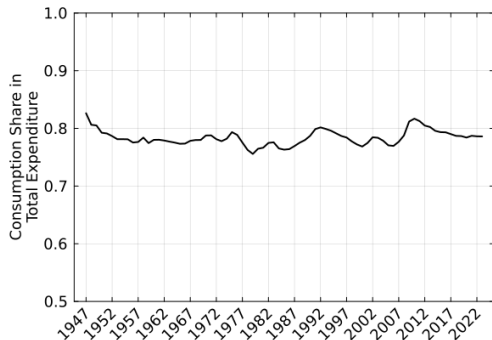
Relative Price of Investment and Aggregate TFP



- ▶ Since 1980:
 - ▶ declining relative price of investment.
 - ▶ rising aggregate TFP, at constant rate.

ST and IBSC

Total and Consumption Expenditure Composition



- ▶ Consumption expenditure roughly stable as a share of GDP.
- ▶ Strong decline in the share of goods in total consumption expenditure.

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Structural Transformation Model

Sectoral Production

- ▶ Two consumption sectors: goods g and services s .
- ▶ The final good can be allocated to consumption expenditure or investment.
- ▶ Sectoral value added is produced with Cobb–Douglas technologies:

$$Y_{j,t} = A_{j,t} K_{j,t}^{\theta} L_{j,t}^{1-\theta}, \quad j \in \{g, s\}.$$

- ▶ Capital and labor are fully mobile across sectors,
- ▶ + firms face common equilibrium factor prices (r_t, w_t) :

$$\rightarrow \frac{K_{g,t}}{L_{g,t}} = \frac{K_{s,t}}{L_{s,t}}$$

Structural Transformation Model

Investment Technology

- ▶ Investment is produced using goods and services as intermediate inputs:

$$I_t = A_{x,t} \left[\omega^{1/\varepsilon} X_{g,t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} X_{s,t}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}.$$

- ▶ $X_{g,t}$ and $X_{s,t}$ are goods and services used in investment production.
- ▶ $A_{x,t}$ captures investment-specific technical change.
- ▶ Capital evolves according to:

$$\dot{K}_t = I_t - \delta K_t.$$

Equilibrium Prices

Model in Investment Units

- ▶ Investment good as the numeraire.
- ▶ Equilibrium relative prices are pinned down by sectoral productivities.

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}.$$

- ▶ Prices relative to investment are:

$$P_{g,t} = \frac{\mathcal{A}_t}{A_{g,t}}, \quad P_{s,t} = \frac{\mathcal{A}_t}{A_{s,t}},$$

where

$$\mathcal{A}_t = A_{x,t} \left[\omega A_{g,t}^{\varepsilon-1} + (1 - \omega) A_{s,t}^{\varepsilon-1} \right]^{\frac{1}{\varepsilon-1}}.$$

Aggregate Technology Representation

Aggregation Result

- ▶ Aggregate final output, measured in units of investment, is:

$$Y_t = P_{g,t}C_{g,t} + P_{s,t}C_{s,t} + I_t.$$

- ▶ Since factor ratios are equalized across sectors:

$$Y_t = \mathcal{A}_t K_t^\theta L_t^{1-\theta}.$$

$\mathcal{A}_t$ is aggregate TFP measured in units of investment.

Structural transformation is consistent with an aggregate production function.

Preferences and Structural Transformation

PIGL Demand System

- Preferences over g and s are represented by the PIGL indirect utility function:

$$V(e_t, P_{g,t}, P_{s,t}) = \frac{1}{\chi} \left[\left(\frac{e_t}{P_{s,t}} \right)^{\chi} - 1 \right] + \frac{\eta}{\gamma} \left[1 - \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} \right],$$

with

$$\eta > 0, \quad 1 > \gamma > \chi > 0.$$

- Goods share in total consumption expenditure:

$$s_{g,t} = \frac{P_{g,t} c_{g,t}}{e_t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma}.$$

Aggregate Balanced Growth Path

ABGP: BGP for aggregate variables + ST within Consumption and Investment

- ▶ Assume sectoral productivities grow at constant rates:

$$A_{g,t} = A_{g,0}e^{g_g t}, \quad A_{s,t} = A_{s,0}e^{g_s t}, \quad \mathcal{A}_t = \mathcal{A}_0e^{g_{\mathcal{A}} t},$$

with

$$g_{\mathcal{A}} > g_g > g_s.$$

- ▶ Along the ABGP, aggregate per-capita variables grow at:

$$g_k = g_e = g_y = \frac{g_{\mathcal{A}}}{1 - \theta} + g_h.$$

- ▶ The model features aggregate growth when investment is the numeraire.
- ▶ But relative prices and expenditure shares still change over time.

Divisia Growth Along the ABGP

Real GDP Growth

- ▶ Along the ABGP, the Divisia growth rate of real GDP is:

$$g_t^D = s_e \underbrace{[s_{g,t}g_{c_g} + (1 - s_{g,t})g_{c_s,t}]}_{\text{Growth rate of real consumption expenditure}} + (1 - s_e)g_x.$$

- ▶ s_e : share of consumption expenditure in GDP.
 - ▶ g_x : is the growth rate of investment.
 - ▶ $s_{g,t}$: share of goods in consumption expenditure.
- ▶ Since $s_{g,t}$ and $g_{c_s,t}$ change over time, and $g_{c_g} > g_{c_s,t}$:
 $\implies g_t^D$ declines along the ABGP.

The Divisia growth rate differs from the growth rate of GDP measured in investment units.

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Bellman Representation of Preferences

Non-Homothetic and Time-Varying Preferences

- ▶ The Bellman representation of preferences is:

$$W_t(c_g, c_s, x) = U(c_g, c_s) + \nu_t x.$$

- ▶ ν_t : marginal value of capital: $\nu_t = v'(k_t)$.
- ▶ The Bellman representation is:
 - ▶ Quasi-linear $\rightarrow$ Non-Homothetic.
 - ▶ Time-varying $\rightarrow \nu_t$ changes over time.

The welfare ordering used to value current output changes along the growth path.

Bellman Representation of Preferences

Indirect Utility and Expenditure Functions

- ▶ In equilibrium, consumption expenditure is:

$$e_t = (\nu_t P_{s,t}^\chi)^{\frac{1}{\chi-1}}.$$

- ▶ Given net income m and prices (P_g, P_s) , the indirect utility function is

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu \left[m - (\nu P_s^\chi)^{\frac{1}{\chi-1}} \right].$$

- ▶ The associated expenditure function is

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}.$$

- ▶ Quasi-linearity in investment makes income enter linearly through νm .
- ▶ The welfare correction depends on prices and the preference order indexed by ν .

Equivalent variation Measure of Income

- ▶ Fisher-Shell Principle: Welfare comparisons require a common preference ordering.
- ▶ Let
 - ▶ t : date of the preference order,
 - ▶ τ : date whose welfare is evaluated,
 - ▶ z : date of reference prices.

$$\widehat{m}_{t,z,\tau} = f(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t).$$

- ▶ $\widehat{m}_{t,z,\tau}$: income at prices z needed to attain the welfare of period τ , using preferences of t .

Proposition 3

Equivalent Variation vs Current Income

- Under the Bellman representation,

$$\hat{m}_{t,z,\tau} = m_{\tau} + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} = \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right).$$

- The EV measure equals observed income plus a price/preference correction.
- The correction depends on hypothetical consumption reallocation gains under the base preference order.

t → preference date τ → welfare comparison date z → reference price date

Proposition 4

Growth Rate of the Equivalent Variation Measure

- ▶ The growth rate of the EV income measure is

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_{\tau}}{\hat{m}_{t,z,\tau}} \left(g_{\tau}^D + \text{dev}_{t,z,\tau} \right).$$

- ▶ The Divisia growth rate appears as a benchmark:

$$g_{\tau}^D = s_e [s_{g,\tau} g_{c_g} + (1 - s_g) g_{c_s,\tau}] + (1 - s_e) g_x.$$

- ▶ Deviations from Divisia growth arise because fixed-base welfare comparisons imply different expenditure and sectoral shares.

$t \rightarrow$ preference date $\tau \rightarrow$ welfare comparison date $z \rightarrow$ reference price date

Three Welfare-Based Indices

How to choose preferences and prices?

1. Chained Fisher–Shell / Divisia index

$$\mathcal{D}_z = \int_{t_0}^z g_h^D dh.$$

Preferences and prices update continuously.

2. Current-base Fisher–Shell index

$$\mathcal{P}_{t_1,z} = \log \hat{m}_{t_1,t_1,z} - \log \hat{m}_{t_1,t_1,t_0}.$$

All dates evaluated using current preferences and current prices.

3. Reference-base Fisher–Shell index

$$\mathcal{L}_{t_0,z} = \log \hat{m}_{t_0,t_0,z} - \log m_{t_0}.$$

All dates evaluated using reference-period preferences and prices.

Current-Base and Reference-Base Fisher–Shell Indices

Key Implication

- At the base date, fixed-base Fisher–Shell growth coincides with Divisia growth:

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D, \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D.$$

- Away from the base date, the indices diverge:

$$\frac{d\mathcal{P}_{t_1,z}}{dz} < g_z^D < \frac{d\mathcal{L}_{t_0,z}}{dz} \quad \Rightarrow \quad \mathcal{P}_{t_1,z} < \mathcal{D}_z < \mathcal{L}_{t_0,z}.$$

Current-base measures understate past growth.

Reference-base measures overstate it.

t → preference date τ → welfare comparison date z → reference price date

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Calibration

Externally Calibrated Parameters

θ	ρ	δ	n	g_h	g	g_s	g_g	ζ
0.333	0.040	0.080	0.0091	0.0046	0.0130	0.0024	0.0097	0.0137

- ▶ θ, ρ, δ taken from HRV.
- ▶ Extend HRV data through 2023.
- ▶ Compute exponential growth rates of:
 - ▶ population (n),
 - ▶ aggregate labor services (g_h),
 - ▶ sectoral ($\{g_s, g_g\}$) and aggregate (g_A) TFP, via growth accounting.
- ▶ ζ matches the gap between observed and the model-implied relative prices.

Calibration

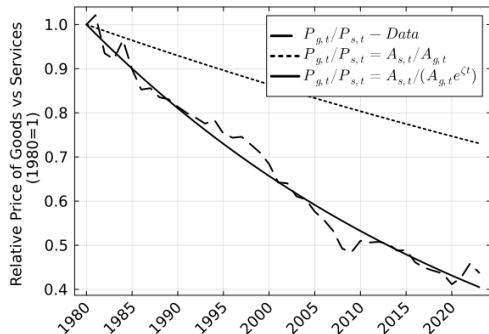
Preference Parameters Estimated via SMM

χ	η	γ
0.360	0.251	0.714

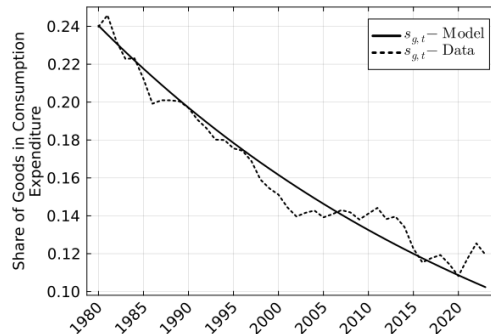
- ▶ η , χ , and γ are estimated by matching the goods' share in consumption expenditure, 1980–2023.
- ▶ The preference system captures both income effects and relative-price effects in consumption.

Calibration Fit

Relative Prices and Expenditure Shares



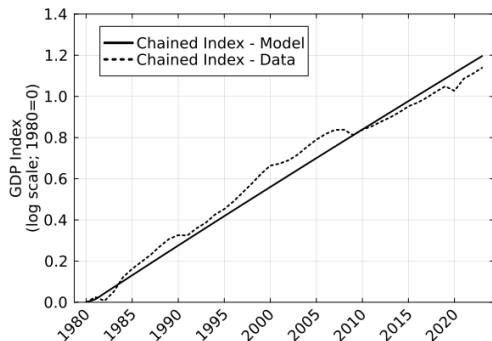
(a) Consumption goods relative price



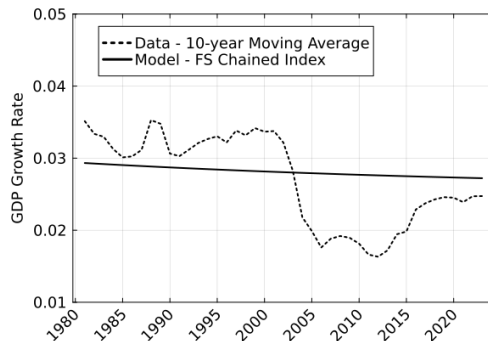
(b) Goods consumption share

Model Fit

GDP Level and Growth



(a) U.S. vs model GDP



(b) U.S. vs model GDP growth

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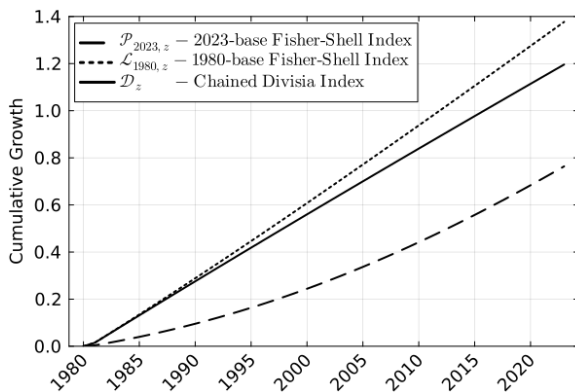
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Welfare-Based Real GDP Measures

Fixed-Base Fisher-Shell Indices vs Chained Divisia Index

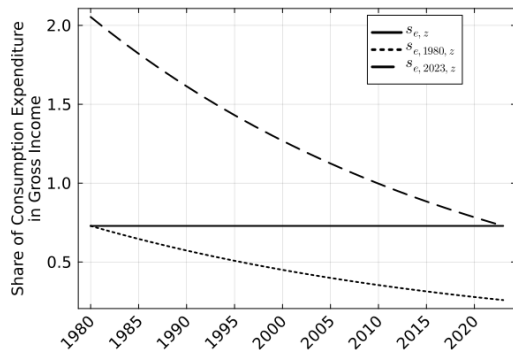


- ▶ The 1980-base Fisher–Shell index lies above the chained Divisia index (+20 p.p. in 2023).
- ▶ The 2023-base Fisher–Shell index lies below it (-23.3 p.p. in 2023).

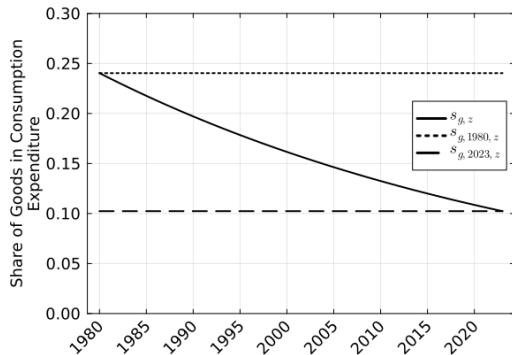
◀ Price-chained version

Why Do the Indices Diverge?

Implied Expenditure Shares



(a) Consumption expenditure share



(b) Goods consumption share

- Fixed-base EV measures evaluate welfare using expenditure shares that differ from the observed equilibrium allocation.

Economic Interpretation

Base-Year Bias

▶ 1980-base Fisher–Shell index

- ▶ uses past preferences to evaluate later allocations;
- ▶ assigns a declining weight to consumption;
- ▶ therefore overweights investment, the faster-growing component;
- ▶ overstates real income growth.

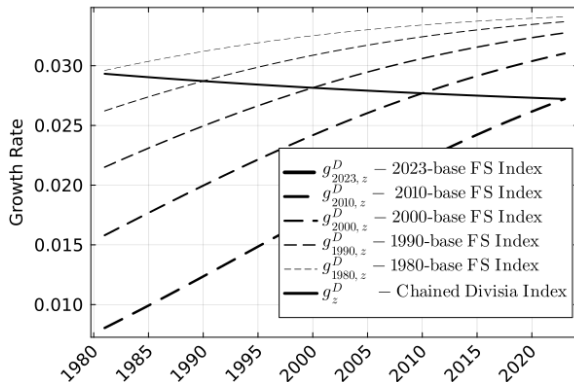
▶ 2023-base Fisher–Shell index

- ▶ uses current preferences to evaluate past allocations;
- ▶ assigns an excessively high weight to consumption in the past;
- ▶ therefore overweights the slower-growing component;
- ▶ understates real income growth.

The chained Divisia index avoids these biases by updating prices and preferences locally.

Current-Time Upgrades

GDP History Revisions



- ▶ Repeated current-base evaluations systematically revise past growth downward.
- ▶ The chained Divisia index delivers a stable welfare-consistent growth path.

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Conclusion

- ▶ Structural transformation and investment-specific technical change:
 - ▶ alter relative prices,
 - ▶ expenditure shares,
 - ▶ and the composition of output.
- ▶ Under these conditions:
 - ▶ welfare comparisons depend on the choice of preference base,
 - ▶ making fixed-base quantity indices systematically biased.
- ▶ The chained Divisia index:
 - ▶ continuously updates prices and expenditure weights,
 - ▶ coincides with the chained Fisher–Shell index in continuous time,
 - ▶ and provides a welfare-consistent measure of real income growth.
- ▶ Quantitatively:
 - ▶ fixed-base measures imply large and growing distortions,
 - ▶ and generate systematic revisions of historical growth.

Thank You



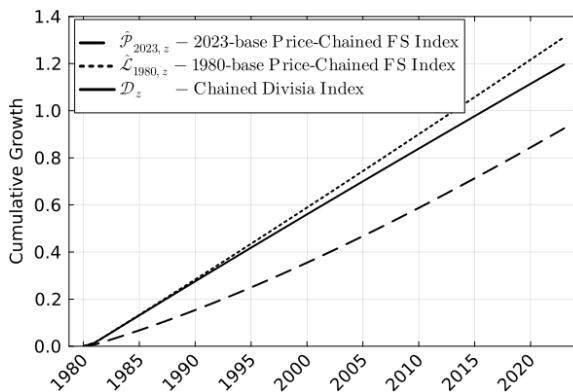
Replication files and online appendix available at:

<https://github.com/jivizcaino/Chained-Indices-Unchained>

Appendix

Appendix

Price-Chained Fisher–Shell Indices

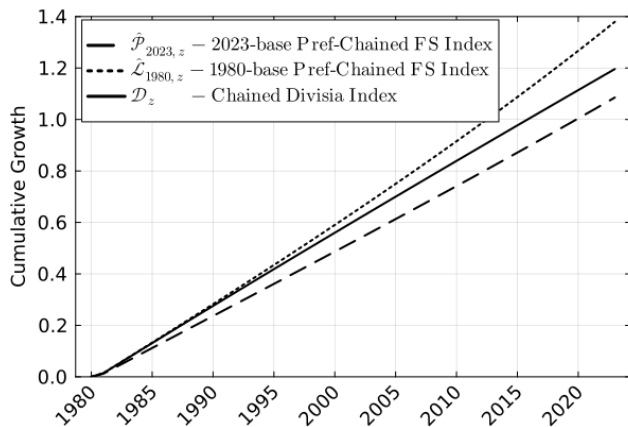


- Prices update continuously, but preferences are kept fixed.
- Most of the discrepancy with the Divisia index comes from holding preferences fixed.

◀ Back to main result

Appendix

Preference-Chained Fisher–Shell Indices



Appendix

Transformation Distortions in Consumption

- ▶ Household expenditure prices differ from producer prices:

$$P_{j,t}^{\tau} = e^{\tau_j t} P_{j,t}, \quad j \in \{g, s\}.$$

- ▶ Producer prices and investment decisions are unchanged:

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}.$$

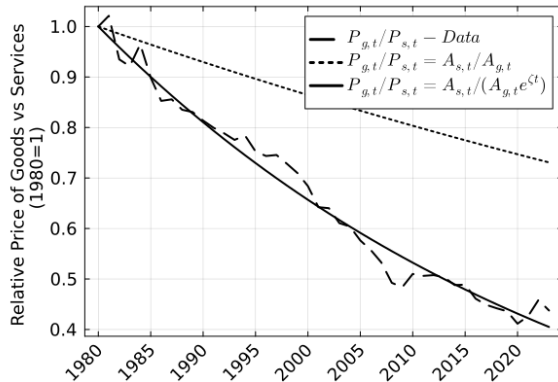
- ▶ Household relative expenditure prices become

$$\frac{P_{g,t}^{\tau}}{P_{s,t}^{\tau}} = \frac{A_{s,t}}{A_{g,t}} e^{(\tau_g - \tau_s)t}.$$

The wedge affects consumption allocation, but not production decisions, factor prices, or the aggregate technology representation.

Appendix

Transformation Wedge and Relative Prices



Appendix

Non-homothetic CES Preferences

- Replace PIGL demand with non-homothetic CES consumption:

$$\omega_c^{1/\sigma} \left(\frac{c_g}{C^{\eta_m}} \right)^{\frac{\sigma-1}{\sigma}} + (1 - \omega_c)^{1/\sigma} \left(\frac{c_s}{C^{\eta_s}} \right)^{\frac{\sigma-1}{\sigma}} = 1.$$

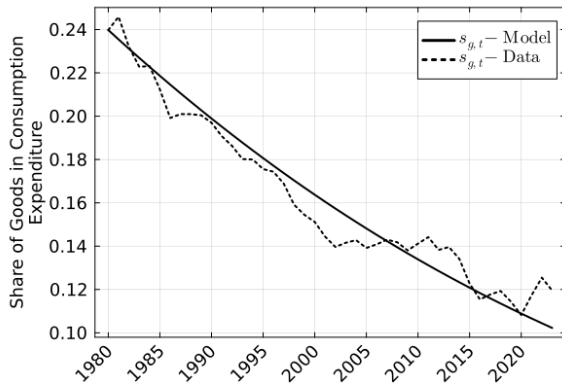
- The expenditure function is available in closed form:

$$E(P_g, P_s, C) = \left[\omega_c P_g^{1-\sigma} C^{(1-\sigma)\eta_m} + (1 - \omega_c) P_s^{1-\sigma} C^{(1-\sigma)\eta_s} \right]^{\frac{1}{1-\sigma}}.$$

- The production side, investment block, factor prices, and aggregation result are unchanged.

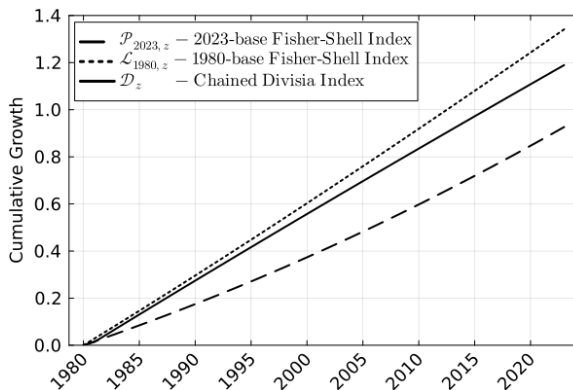
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NHCES Calibration Fit



Appendix

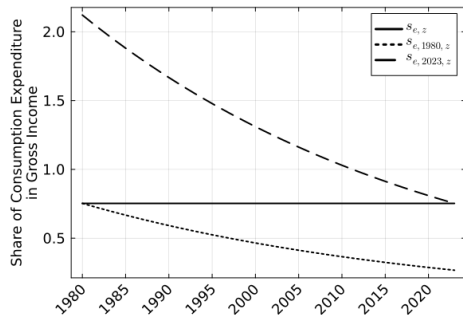
Welfare-based GDP measures under NHCES



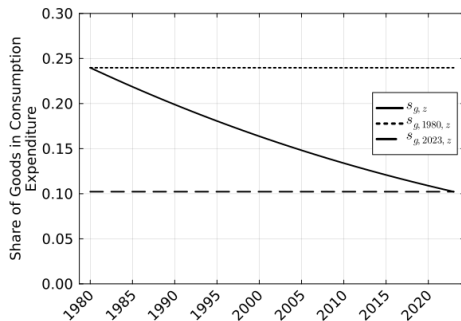
- The same qualitative pattern holds: earlier-base measures lie above Divisia; later-base measures lie below it.

Appendix

Implied Expenditure Shares Under NHCES Preferences.



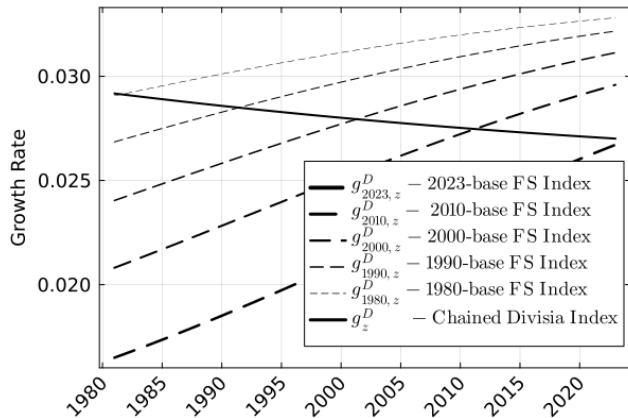
(a) Consumption expenditure share



(b) Goods consumption share

Appendix

NHCES Revisions



Appendix

Indices for Real Consumption Expenditure

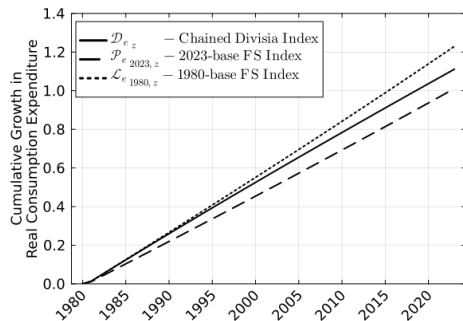
- In the endowment-economy analogue, the fixed-base Fisher–Shell growth rate for real consumption expenditure is

$$g_{e,t,z} = \frac{d \log \hat{e}_{t,z}}{dz} = g_{e,z}^D \left(\frac{e_z}{\hat{e}_{t,z}} \frac{P_{s,t}}{P_{s,z}} \right)^\chi.$$

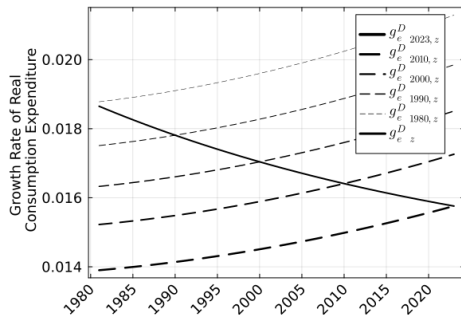
- A chained Fisher–Shell index again coincides with the chained Divisia index.
- Fixed-base indices still produce systematic base-year distortions under non-homotheticity.

Appendix

Real Consumption Expenditure Indices and Revisions



(a) Real consumption expenditure



(b) Growth-rate revisions

Appendix

Model fit Under NHCES

